Project Central

AI use statement

I used Claude (Sonnet 5) to help me brainstorm project ideas for this class using the material we cover. I chose a project that mirrors the structure I use in other courses and Claude helped map different checkpoints to different weeks. Claude also drafted this page.

Checkpoints

The checkpoints build, step by step, the work you need for the final project, so we complete the work over the quarter instead of all at the end. **If you receive feedback on something in a checkpoint, you are expected to fix it for the next checkpoint and the final project.**

Checkpoint Instructions	Points	Editing file	Rubric	Due Date	Topics
Checkpoint 0	16			10/11	Choose your scenario + find input values in your source packet
Checkpoint 1	12			10/18	Conditional probability + Bayes' rule: when is a positive really positive?

Checkpoint Instructions	Points	Editing file	Rubric	Due Date	Topics
Checkpoint 2	12			11/15	Discrete random variables: counting events
Checkpoint 3	12			11/29	Continuous random variables + the Central Limit Theorem

All checkpoints are submitted as a **rendered Quarto HTML file** (with `code-fold: true`) plus your `.qmd` file.

Project Overview: Probability in the Wild

This quarter, you will follow **one public health scenario**. Each time we learn a new set of probability and simulation tools, you will apply them to your scenario in a checkpoint. At the end of the quarter, you will pull your checkpoints together into a short brief written for the public health program in your scenario.

Every checkpoint follows the same three steps:

1. **Solve it:** Set up the problem with probability notation and solve it analytically (by hand or with R as a calculator).
2. **Simulate it:** Write a simulation in R and check that it agrees with your analytic answer. If the two answers disagree, explain why.
3. **Say it:** Interpret your results in plain language for someone who is not a statistician.

You may work **alone or in pairs**. If you work in a pair, you keep the same partner and scenario for the whole quarter.

Scenario menu

Pick one of the three scenarios below. Each one comes with a fixed setting and a short packet of published sources, so you do not need to search for sources yourself. The full scenario cards are in Checkpoint 0. You may propose your own scenario only with my approval (see Checkpoint 0).

Scenario	CP 1: Conditional probability & Bayes	CP 2: Discrete RV	CP 3: Continuous RV & CLT
A. Community breast cancer screening program	P(cancer | abnormal mammogram)	Follow-up appointments needed after a van day of 40 screens	Size of a tumor found by screening
B. County COVID-19 testing and contact tracing	P(infected | positive rapid test)	Household contacts of a case who get infected	Incubation period
C. State newborn heart screening program	P(heart defect | positive pulse oximetry screen)	Babies born with a heart defect in the state in a year	Birth weight

> **Use real numbers**
>
> Every input value (prevalence, sensitivity, rate, mean, etc.) comes from your scenario's source packet or from a course value on your scenario card. Cite the source each time you use a value. You do not need to analyze a dataset. The data in this project come from simulation.

Final Project: Public Health Brief

The final project is due 12/11.

In the final project, you will revise your three checkpoints based on feedback and combine them into one story about your scenario. You will submit:

1. **Technical appendix:** Your revised Checkpoints 1–3 in a single rendered Quarto file.
2. **Public health brief:** A 2-page brief (or a 5-minute recorded explanation) for the manager of the public health program in your scenario. It should answer: *What does probability tell us about this program, and what should the manager keep in mind when making decisions?*

The Final Project Instructions describe both pieces, what each section of the brief should include, and the rubric.

Final Project Instructions